

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN ANIMALS (SUB-STRAND 3.2)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger? Think back to the empty, split skin (exuvia) on the maize leaf in the school garden, lying beside a larger newly emerged grasshopper or cicada. Use what you observed and learned in Lessons 1 to 5 to answer each section below. Write in full sentences, use correct scientific words (for example exoskeleton, moulting/ecdysis, metamorphosis, Brain Hormone, Ecdysone, Juvenile Hormone), and where asked, give a Claim, Evidence and Reasoning. You may draw and label diagrams or sketch graphs in the space under each prompt. Only describe patterns you actually saw in our class investigations.

Part 1: Outgrowing the Shell (Stepwise Growth)

Look back at the exuvia on the maize leaf beside the larger grasshopper or cicada. (a) Explain how the insect became bigger than the hard skin it was wearing. In your answer, say what the exoskeleton can and cannot do, and when in the moulting process the insect actually increases in size. (b) Describe the shape of the graph of nymph length against time that we plotted in class, and explain why it looks like that. Use the words moulting (ecdysis) and exuvia.

Write your answer here:

Part 2: Two Growth Patterns Compared

In class we plotted data for a Kenyan broiler chick, a child, and a grasshopper nymph. (a) Describe the shape of the growth curve for the chick and child, and name the four phases you identified. (b) State the formula for growth rate that we used and say in which phase growth rate is highest and in which phase it is close to zero. (c) Explain why the chick's graph looks different from the grasshopper nymph's graph.

Write your answer here:

Part 3: The Hormone Control System

Moulting and metamorphosis are not random. Using the experiments on larvae that you analysed (removing the brain/neurosecretory cells, injecting Ecdysone, adding extra Juvenile Hormone, removing the corpora allata early), write a CER statement answering: What controls when an insect moults and what kind of moult it has? Include the role of Brain Hormone, Ecdysone and Juvenile Hormone in your reasoning.

Write your answer here:

Part 4: Two Pathways of Metamorphosis

Compare a grasshopper and a butterfly (or housefly). (a) Name the stages in each life cycle and say which is incomplete and which is complete metamorphosis. (b) Using Juvenile Hormone, explain why grasshopper nymphs look like small adults but a caterpillar looks nothing like the adult butterfly. (c) Give TWO ways that having separate larval and adult stages might help an insect survive.

Write your answer here:

Part 5: Putting It All Together (Final CER)

Write a complete CER answer to the Driving Question: How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger? Use the exuvia on the maize leaf, a growth curve, and the hormones in your evidence. Then, in one or two sentences, suggest a new question you still have about this topic.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining stepwise growth and the exuvia	Clearly explains that the exoskeleton cannot stretch, a new larger cuticle forms beneath, the old one splits to leave the exuvia, and the insect expands while the new cuticle is soft. Links this accurately to the stepped graph.	Explains that the insect moults and gets bigger afterwards, and describes the stepped graph, but leaves out one detail such as the new cuticle forming first or the soft cuticle expanding.	States that the insect sheds its skin but does not explain why it must moult or how it becomes bigger, or treats the exuvia like peeling skin.
Comparing growth curves (sigmoid vs intermittent)	Correctly names all four phases of the S-curve, uses the growth rate formula, identifies where rate is highest and lowest, and explains the difference from the insect staircase using the rigid exoskeleton.	Describes both curve shapes and names most phases or the formula, with a reasonable but incomplete explanation of the difference.	Describes only one curve or confuses the shapes, with few or incorrect phases and no clear link to the exoskeleton.
Hormonal control (Brain Hormone, Ecdysone, JH)	Accurately explains the sequence from Brain Hormone to prothoracic glands to Ecdysone, and the role of high, falling and absent JH, using specific experimental evidence in a full CER.	Names the three hormones and their main roles and uses some experimental evidence, with minor gaps in the reasoning.	Names hormones but mixes up their roles, or gives claims with little or no evidence or reasoning.
Metamorphosis pathways and survival value	Gives correct stages of both pathways, explains with JH why shapes differ, and gives two clear survival advantages.	Gives correct stages and a mostly accurate explanation of the shape difference, with one survival advantage.	Lists stages with errors or does not link the difference to JH or survival.
Final CER and scientific communication	A complete, well-organised Claim, Evidence and Reasoning that answers both parts of the Driving Question using class evidence, correct vocabulary, and a thoughtful new question.	A mostly complete CER that answers the Driving Question with some class evidence and mostly correct vocabulary.	An incomplete CER that answers only part of the Driving Question, with little evidence and limited scientific vocabulary.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided

appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.
